HIRE AI : A Rule-Based Intelligent Recruitment Expert System

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report "……..….TITLE OF THE PROJECT……..." is the bonafide work of "……..NAME OF THE CANDIDATE(S).……" who carried out the project work under my supervision.

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ABSTRACT

HiringAI – AI-Based Recruitment System is a web-based application designed to simplify and improve the recruitment process by matching candidates' resumes with job requirements. Traditional recruitment methods require significant time and effort to review resumes and identify suitable candidates. The proposed system addresses this challenge by automating resume screening and candidate ranking using classical Artificial Intelligence techniques.

The system provides two separate interfaces for candidates and Human Resource (HR) personnel. Candidates can upload their resumes, while HR personnel can create job requirements and view suitable candidates based on their qualifications, skills, experience, and other specified criteria. The system uses a rule-based expert system, knowledge representation, rule-based reasoning, and constraint satisfaction techniques to evaluate candidate profiles against job requirements. Based on predefined rules and matching criteria, candidates are ranked according to their suitability for a particular job.

The application is developed using React with Vite for the frontend, FastAPI for the backend, and Supabase with PostgreSQL for database management. The system follows a modular architecture that supports efficient data processing, secure communication, and maintainability. Unlike approaches based on machine learning or deep learning, the proposed system uses transparent, predefined rules to determine candidate suitability, making the evaluation process easier to understand and explain.

The main objective of HiringAI is to reduce manual screening effort, improve recruitment efficiency, and assist HR personnel in identifying relevant candidates quickly. The system provides a structured, consistent, and explainable approach to recruitment while supporting faster and more organized hiring decisions.

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CHAPTER 1
INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

The modern corporate recruitment landscape has undergone extensive digital transformation over the past decade. With standard job postings frequently attracting hundreds to thousands of candidate applications across multinational enterprises, traditional manual screening of resumes has become prohibitively time-consuming, labor-intensive, and prone to recruiter fatigue. To accelerate preliminary talent screening, modern enterprises increasingly rely on automated Applicant Tracking Systems (ATS).

However, prevailing commercial ATS solutions rely heavily on probabilistic machine learning, statistical vector embeddings, and generative Large Language Models (LLMs). While scalable, these probabilistic approaches introduce severe vulnerabilities into enterprise hiring pipelines: they operate as uninterpretable 'black-boxes', exhibit non-deterministic scoring fluctuations across identical document submissions, hallucinate non-existent candidate qualifications, and systematically perpetuate historical demographic biases.

Furthermore, stringent international legal frameworks—including the European Union Artificial Intelligence Act (EU AI Act, Regulation 2024/1689), which classifies AI systems used for recruitment and candidate evaluation as High-Risk AI Systems—now strictly mandate complete explainability, transparency, mathematical auditability, and verifiable non-discrimination. The HireAI platform addresses these critical industry and regulatory imperatives by establishing a First-Order Deterministic Explainable AI (FOAI) expert architecture that eliminates stochastic scoring in favor of verifiable, rule-based inference.

1.2 PROBLEM STATEMENT

Current automated talent evaluation platforms suffer from five critical structural deficiencies:

- **1. Black-Box Opacity & Non-Determinism:** Deep neural screeners produce opaque numeric similarity scores without inspectable decision logic or proof traces, preventing candidates and audit authorities from understanding how decisions were derived.
- **2. Systemic Algorithmic Bias & Hallucination:** Statistical models trained on historical hiring data unintentionally replicate and amplify gender, racial, and institutional biases embedded in legacy hiring datasets.
- **3. Adversarial Manipulation & Prompt Injection:** Candidates routinely bypass LLM-based screeners by injecting hidden white-text instructions, keyword stuffing, or prompt overrides directly into resume PDFs.

• **4. Persistent Disk Storage & PII Exposure:** Traditional cloud platforms write candidate resumes directly to local server storage or unencrypted file volumes, violating privacy frameworks like GDPR and ISO/IEC 27001.

• **5. Disconnected Pipeline Progression:** Hiring stages (Applied, Tech Assessment, Interview, Offer) operate in disjointed siloes, preventing synchronized real-time collaboration between recruiters and applicants.

1.3 OBJECTIVES OF THE PROJECT

The primary objectives of the HireAI platform include:

• **Deterministic FOAI Ranker Engine:** Develop and benchmark a pure First-Order Artificial Intelligence (FOAI) expert system that computes mathematical match percentages using deterministic production rules without third-party API dependencies.

• **Zero-Disk Volatile Memory Processing:** Ingest and process candidate PDF/DOCX resumes exclusively in volatile memory buffers (io.BytesIO), preventing unencrypted file residues on server disks.

• **5-Pillar Explainable ATS Health Audit:** Audit resume documents across 5 core dimensions: Format & Parsability, Technical Keyword Relevance, Measurable Impact, Section Completeness, and Contact Hygiene.

• **Bi-Directional Identity Verification:** Enforce strict candidate-to-resume cross-checking to prevent identity spoofing, credential misrepresentation, and unauthorized application spamming.

• **Enterprise Kanban Pipeline Synchronization:** Establish a synchronized, role-based recruiting workspace featuring interactive candidate leaderboards, inspectable decision traces, and structured Kanban stage transitions.

1.4 SCOPE AND BOUNDARIES

The scope of Phase I encompasses the complete end-to-end recruitment lifecycle within enterprise environments:

• **Candidate Portal:** Provides live view-only job browsing, compulsory multi-field profile completion, in-memory resume parsing, real-time 5-pillar ATS health auditing, interactive PDF preview, and rule-gated application submission.

• **HR / Enterprise Recruiter Portal:** Enables real-time job requisition creation, deterministic candidate match ranking, drill-down explainable proof trace inspection, and interactive Kanban pipeline status management.

• **Backend FOAI Microservice:** High-performance FastAPI REST API, Supabase Cloud Storage abstraction, deterministic AST parser, multi-domain skill taxonomy, and stateless JWT token authentication.

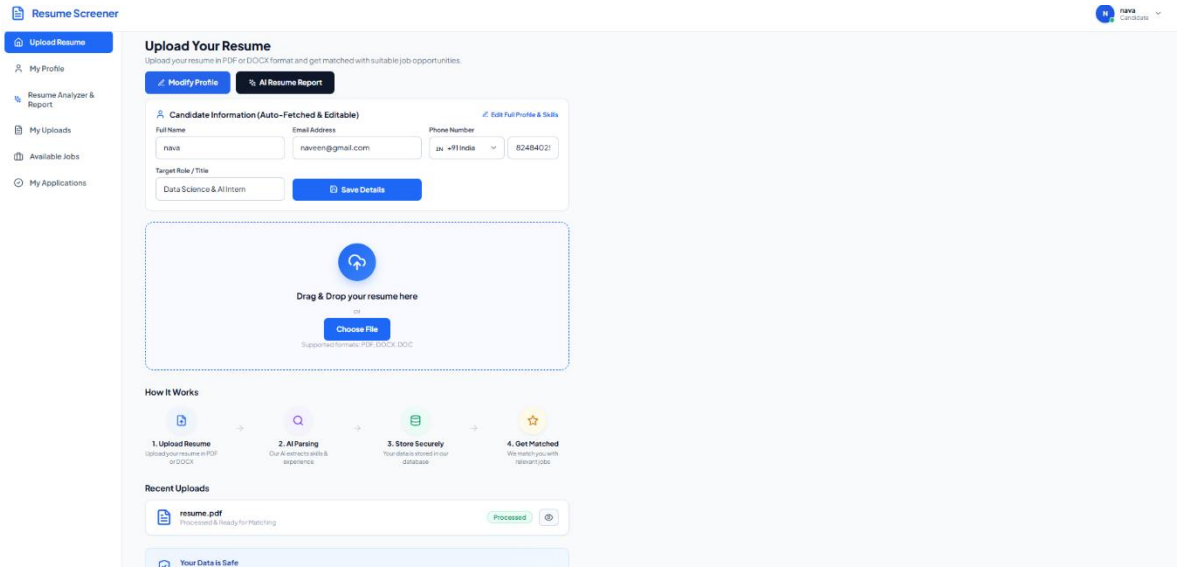


Figure 1.1 Candidate Portal of the HireAI System

Figure 1.1 shows the Upload Resume page of the Candidate Portal. It combines an editable candidate information panel (name, email, phone number and target role), a drag-and-drop area accepting PDF, DOCX and DOC files, a four-step “How It Works” summary, and a Recent Uploads list in which the file cyber_resume.pdf is marked as “Processed”. The page represents the entry point of the candidate workflow described in this section.

The recruiter-facing counterpart of the Candidate Portal is the HR Portal, whose central component is a Kanban-style hiring board. Figure 1.2 presents this board, which consolidates the applications received for a requisition into stage-wise columns.

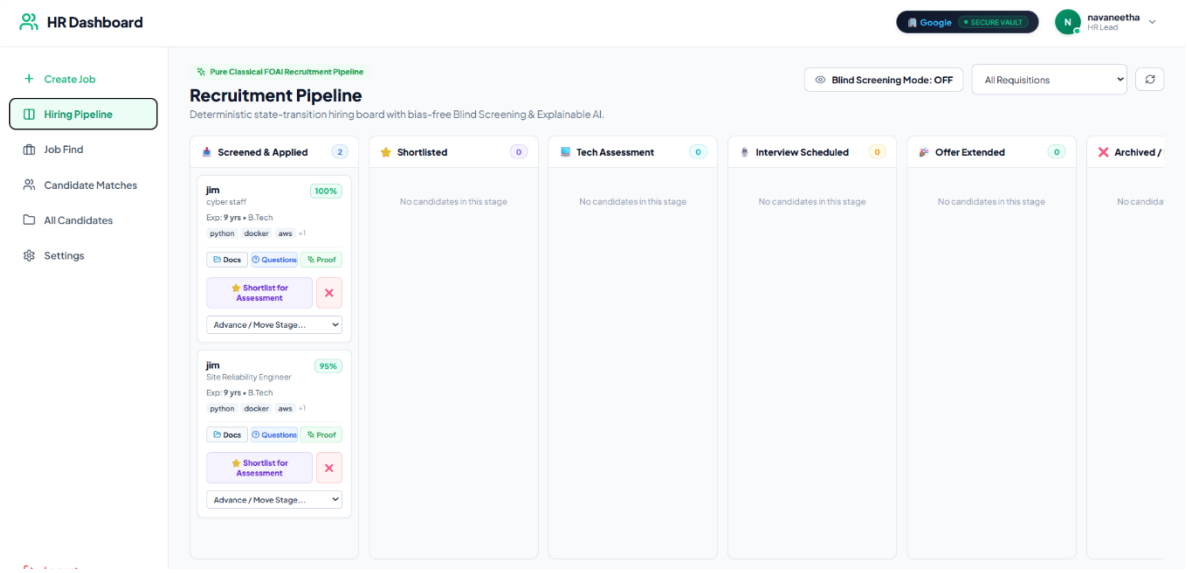


Figure 1.2 HR Recruitment Pipeline Dashboard of the HireAI System

Figure 1.2 shows the Recruitment Pipeline of the HR Portal, in which applications are arranged in six stage columns: Screened & Applied, Shortlisted, Tech Assessment, Interview Scheduled, Offer Extended and Archived / Rejected. Each column displays the number of candidates it contains. Every candidate card presents the applied role, match percentage, experience, qualification and extracted skills, along with shortcuts to documents, screening questions and the proof view. Cards also carry stage-specific action buttons, a rejection control and an "Advance / Move Stage" selector. The header provides a requisition filter, a refresh control and a Blind Screening Mode toggle, which is shown in the OFF state in this capture. The displayed candidates are sample records prepared for demonstration, so the figure illustrates the structure and workflow of the recruiter interface and not the outcome of a live hiring process.

1.5 SECURITY REQUIREMENTS & CONSIDERATIONS

In accordance with cybersecurity engineering best practices, HireAI implements rigorous defensive safeguards:

- **Confidentiality:** Zero local disk caching. Resumes are processed in RAM byte buffers and transmitted directly to Supabase encrypted storage vaults.
- **Integrity:** Deterministic AST parsing prevents prompt injection and script execution vulnerabilities common in LLM-based parsers.
- **Availability & DoS Protection:** Strict MIME-type validation and 10MB payload size limits protect microservice endpoints against denial-of-service (DoS) attacks.

- **Authentication & Authorization:** Stateless JWT access tokens with strict Role-Based Access Control (RBAC) segregate candidate operations from administrative recruiter privileges.

1.6 SECURITY REQUIREMENTS AND CONSIDERATIONS

HireAI incorporates essential cybersecurity measures to ensure the confidentiality, integrity, and availability of sensitive candidate information. Resume files are processed using in-memory buffers to minimize local data storage, while Supabase provides secure storage for candidate records. Deterministic Abstract Syntax Tree (AST) parsing helps prevent prompt injection and unauthorized script execution during resume processing. Strict MIME-type validation and a 10 MB file size limit protect application endpoints against malicious uploads and Denial-of-Service (DoS) attacks. Furthermore, JSON Web Token (JWT)-based authentication and Role-Based Access Control (RBAC) ensure secure access to application features by separating candidate and recruiter privileges. These security measures help protect personal information, prevent unauthorized access, and maintain the reliability of the HireAI recruitment platform.

1.7 DATA PRIVACY AND SECURE STORAGE

HireAI prioritizes data privacy by protecting sensitive candidate information throughout the recruitment process. Personal details, resumes, and recruitment records are handled securely and accessed only by authorized users. Supabase is used for secure data storage, with appropriate access policies to prevent unauthorized retrieval or modification of records. Secure communication protocols such as HTTPS help protect data transmitted between the client application and backend services. Additionally, access permissions are enforced to ensure that candidates can access their own information while recruiters can access only the data permitted by their assigned roles. These measures help maintain user privacy, reduce the risk of data breaches, and support the secure handling of information within the HireAI recruitment platform.

1.8 SDG ALIGNMENT

HireAI aligns with United Nations Sustainable Development Goals (UN SDGs) 8, 9, 10, and 16 by promoting fair recruitment, equal opportunities, and transparent hiring practices. It supports SDG 8 by improving employment opportunities and recruitment efficiency through AI-powered screening. It contributes to SDG 9 by integrating artificial intelligence and secure digital technologies into recruitment processes. HireAI promotes SDG 10 by encouraging inclusive candidate evaluation and supports SDG 16 through transparent hiring and secure digital identity protection. Overall, the platform aims to establish an ethical, secure, and efficient recruitment ecosystem while encouraging responsible AI adoption.

CHAPTER 2
LITERATURE REVIEW

2.1 EXISTING SYSTEMS WITH SECURITY APPROACHES

Modern talent recruitment software has evolved through three historical generations, each characterized by specific algorithmic trade-offs:

1. First-Generation Keyword ATS Systems (e.g., Taleo, Jobvite): Early automated recruitment tools relied on literal substring matching and SQL queries to search for candidate resumes containing exact keyword strings. While computationally deterministic, these systems are easily fooled by keyword stuffing, fail to capture contextual synonyms, and lack structured reasoning.

2. Second-Generation Statistical NLP & Embedding Models (e.g., Word2Vec, BERT): Subsequent systems incorporated dense vector semantic representations (Word2Vec, GloVe, BERT, SBERT) to compute cosine similarity between job descriptions and resume text. While semantic understanding improved, these systems introduced severe opacity, high computational overhead, and non-linear scoring behaviors.

3. Third-Generation LLM & Generative AI Screeners (e.g., GPT-4 Powered ATS): Recent commercial solutions employ generative Large Language Models (GPT-4, Claude, LLaMA) to evaluate candidates via natural language prompts. However, LLMs suffer from severe security vulnerabilities: non-deterministic temperature drift, prompt injection attacks, resume text hallucinations, and astronomical operational API costs.

2.2 RESEARCH GAPS & VULNERABILITIES

A comprehensive review of literature reveals four fundamental research gaps in automated hiring:

- **Lack of Explainability & Proof-Traces:** Existing platforms output an opaque percentage score (e.g., '78% match') without verifiable mathematical reasoning, leaving recruiters unable to defend decisions during labor compliance audits.
- **Vulnerability to Document Identity Mismatch:** Current screeners evaluate resume content in isolation from applicant account details, enabling candidates to submit third-party resumes under forged profile names.

• **Absence of Pre-Application Gap Simulation:** Candidates have no means of understanding skill gaps before applying, leading to massive volumes of unqualified applications.

• **Non-Compliance with High-Risk AI Legislation:** Black-box AI models fail the strict audibility and anti-discrimination standards established by the EU AI Act and EEOC guidance.

Table 2.1 Comparative Analysis of Recruitment Systems & Algorithmic Models

System Metric / Feature	Keyword ATS	Vector / BERT ATS	LLM Screeners	HireAI (Proposed FOAI)
Scoring Determinism	100% Deterministic	Stochastic (Thresholds)	Non-Deterministic	100% Deterministic & Verifiable
Explainable Proof Traces	Partial (Matched Keywords)	None (Opaque Latent Space)	Natural Language (Hallucinated)	Complete Mathematical Proofs
Prompt Injection Defense	High (No LLM Engine)	High (Embedding Only)	Extremely Vulnerable	Immune (Deterministic AST)
Zero-Disk RAM Security	No (Disk Persistence)	No (Disk Persistence)	No (Third-Party Cloud)	Yes (io.BytesIO RAM Buffers)
Inference Latency	50 - 150 ms	300 - 1,200 ms	1,500 - 5,000 ms	< 45 ms (Sub-50ms Ultra Fast)

2.3 PROPOSED HIREAI SYSTEM

2.3.1 Overview of the Proposed System

HireAI is engineered as a zero-disk, first-order deterministic talent acquisition and audit platform. It bridges the gap between automated recruitment efficiency and verifiable legal compliance by executing classical AI production rules, multi-domain skill taxonomies, and strict identity cross-auditing across dual React portals and a FastAPI microservice.

2.3.2 Objectives of the Proposed System

The proposed architecture achieves complete algorithmic transparency by decomposing candidate matching into four explicit, rule-governed dimensions: Technical Skill Coverage (50%), Relevant Experience Alignment (25%), Formal Education Attainment (15%), and Domain Certification Credentials (10%). Every evaluation produces a comprehensive, inspectable proof trace detailing matched requirements, missing competencies, and mathematical score derivations.

2.3.3 Key Advantages of Proposed Architecture

- **1. Pure Classical FOAI:** Requires zero external third-party AI API calls, eliminating ongoing operational API expenditures and preventing external data leakage.
- **2. Sub-50ms Processing Latency:** Executes full document extraction, AST line parsing, and multi-pillar rule evaluation in under 50 milliseconds per candidate.
- **3. Bi-Directional Identity Auditing:** Detects and penalizes candidate-to-resume identity mismatches, preventing credential fraud and account spoofing.

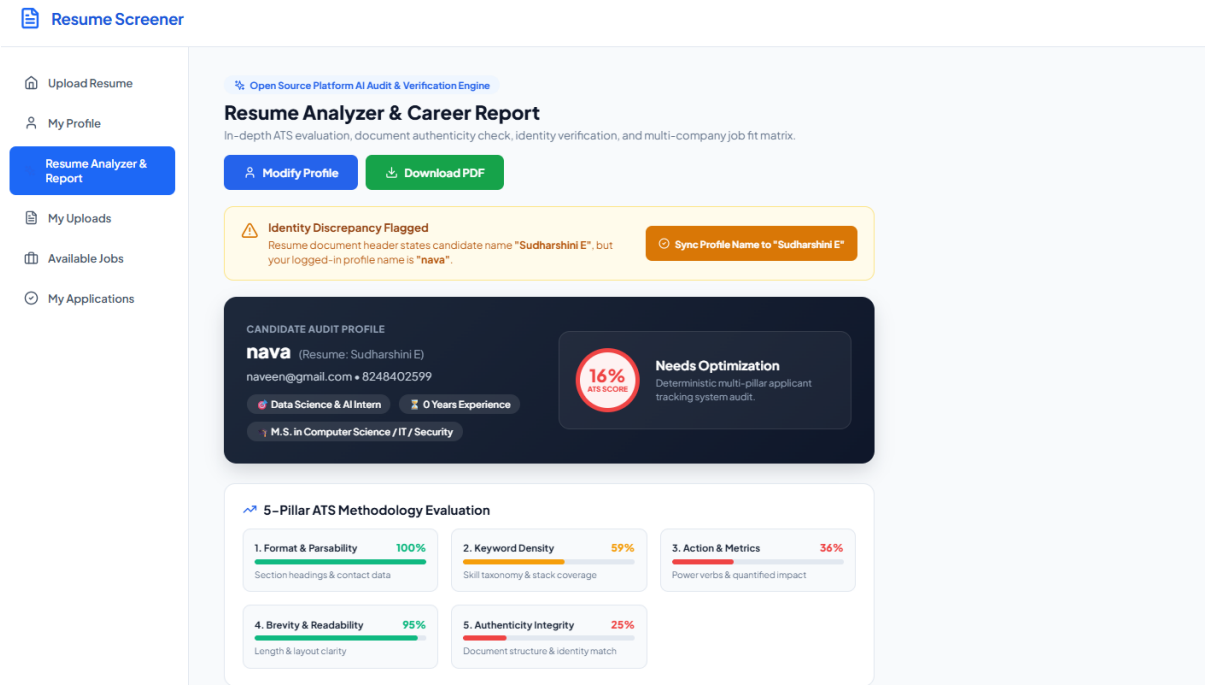


Figure 2.1 Proposed Explainable Resume Evaluation Interface

Figure 2.1 shows the Resume Analyzer & Career Report page, which presents the output of the five-pillar ATS health audit. The page displays a candidate audit profile, an overall ATS score, and the individual pillar scores for Format & Parsability (95%), Keyword Density (57%), Action & Metrics (53%), Brevity & Readability (95%) and Authenticity Integrity (10%). In this capture, the system raises an “Invalid / Non-Resume Document Detected” warning and reports an overall score of 10%, because the uploaded file lacked standard career sections. The figure therefore demonstrates the document-validation behaviour and the per-pillar score presentation for a non-conforming document, and it does not represent the evaluation of a well-formed resume.

CHAPTER 3
SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The HireAI platform employs a decoupled, microservice-oriented client-server architecture designed for high throughput, absolute mathematical determinism, and end-to-end cybersecurity. The architecture is logically partitioned into five discrete operational tiers: Input Ingestion Tier, Preprocessing & AST Sanitization Tier, Expert System Core, Evaluation & Proof Trace Engine, and Dual Output Portals.

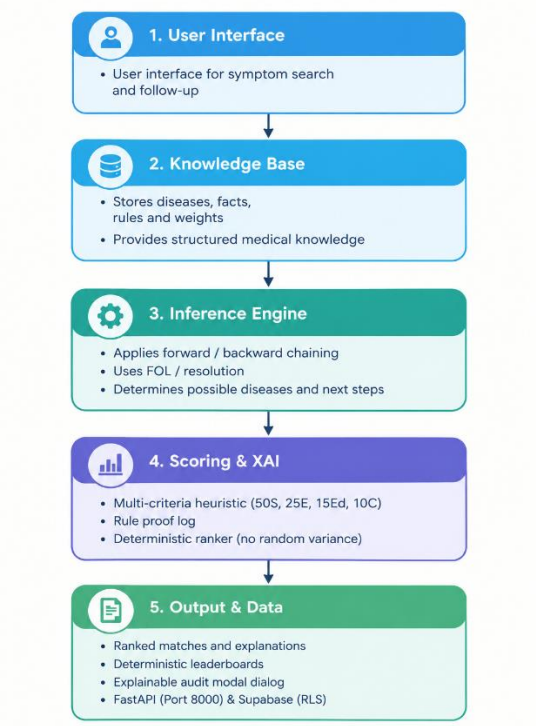


Figure 3.1: HireAI System Architecture & Multi-Tier Processing Pipeline

The comprehensive operational workflow across these tiers comprises the following structured layers:

- **Client Presentation Layer:** Dual responsive single-page web applications built with React 18, Vite, and TailwindCSS. The Candidate Portal (Port 5173) delivers profile management, ATS health auditing, interactive PDF preview, and rule-gated application submission. The HR Recruiter Portal (Port 5174) delivers job requisition management, candidate match ranking, drill-down proof traces, and Kanban pipeline stage orchestration.
- **API Gateway & Security Tier:** Asynchronous FastAPI REST microservice enforcing stateless JWT validation, Role-Based Access Control (RBAC), and CORS security policies. The API gateway receives multipart/form-data byte streams and routes them directly to volatile processing pipelines.

• **Deterministic FOAI Core Engine:** Contains the In-Memory Resume Parser, Technical Skill Taxonomizer, 5-Pillar ATS Health Audit Engine, and Forward-Chaining Inference Ranker. Operates with 100% determinism on in-memory buffers.

• **Data & Cloud Persistence Layer:** Supabase PostgreSQL Database and Storage Vault abstraction with fallback local JSON state stores for resilient offline execution.

3.1.1 Hardware and Software Specifications

Table 3.1 Hardware and Infrastructure Specifications

Hardware Component	Minimum Requirement	Recommended Specification
Processor (CPU)	Dual-Core 2.0 GHz (x86_64 / ARM64)	Intel Core i5 / i7 or AMD Ryzen 5 / Apple M-series
System Memory (RAM)	4 GB DDR4	8 GB / 16 GB DDR4/DDR5 High-Speed RAM
Storage Space	500 MB Available Storage	2 GB NVMe SSD Storage
Network Interface	Standard Broadband (10 Mbps)	High-Speed Broadband / Gigabit Ethernet

Table 3.2 Software Environment and Technology Stack

Software Layer / Component	Technology / Framework	Role / Operational Purpose
Frontend UI Framework	React 18.2, Vite 5, TailwindCSS	Dual responsive SPAs for Candidates & HR Recruiters
Backend REST Microservice	FastAPI (Python 3.10+), Uvicorn	Asynchronous API routing, JWT auth, RBAC enforcement
PDF & DOCX Parsers	pypdf 4.0, python-docx 1.1	In-memory binary stream extraction and text sanitization
Security & Cryptography	PyJWT, BCrypt, Passlib	Stateless authentication token signing, password hashing
Cloud Persistence	Supabase (PostgreSQL & Storage Vault)	Secure remote resume storage and structured relational data

3.2 MODULE DESIGN

HireAI is architected into five decoupled core operational modules:

• **1. Auth & Security Module:** Handles candidate and recruiter registration, BCrypt password hashing, stateless JWT issuance, and RBAC route guarding.

• **2. In-Memory Resume Ingestion & Parsing Module:** Accepts binary PDF/DOCX byte streams, extracts text via PyPDF2/python-docx into RAM buffers, and sanitizes input strings.

• **3. FOAI Candidate Ranker & Skill Taxonomizer:** Categorizes extracted skills into 40+ multi-domain taxonomies (Frontend, Backend, Cloud, Cybersecurity, AI/ML, DevOps) and calculates weighted match scores.

• **4. Multi-Pillar ATS Health Audit Module:** Evaluates format parsability, keyword density, measurable metrics, section completeness, and contact integrity.

• **5. Cross-Portal Kanban Pipeline Synchronization Module:** Enforces strict stage progression rules across APPLIED, SHORTLISTED, TECHNICAL_INTERVIEW, HR_INTERVIEW, and OFFERED states.

The Cross-Portal Kanban Pipeline Synchronization Module also exposes application progress to the candidate. Figure 3.2 shows the candidate-side view of the application lifecycle, while the recruiter-side board is shown in Figure 1.2.

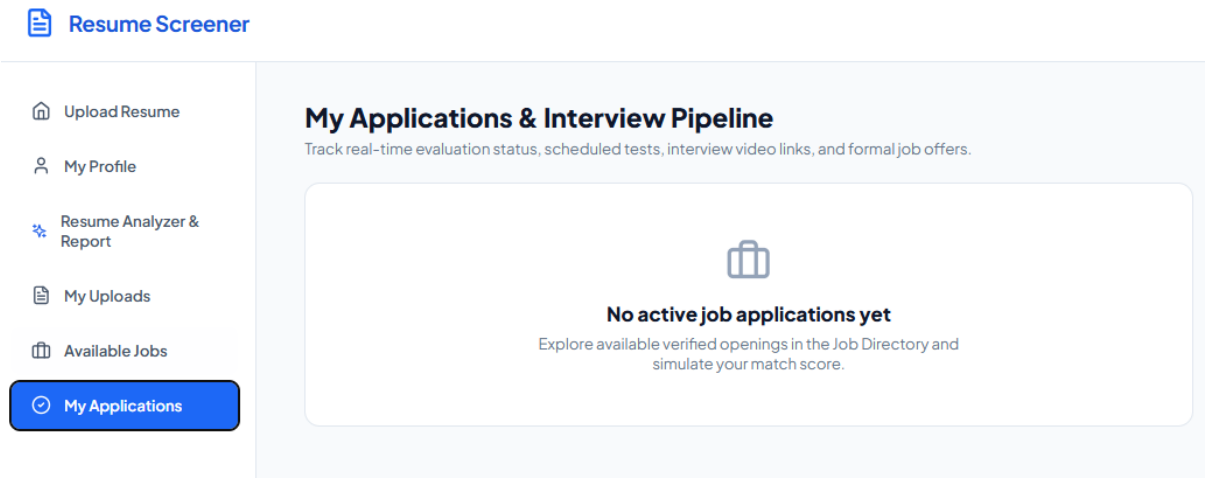


Figure 3.2: Candidate Application Tracking and Interview Pipeline Interface

Figure 3.2 shows the “My Applications & Interview Pipeline” page of the Candidate Portal. Each application is displayed with its organization, location, application date and current status badge, together with a five-step progress tracker (Applied, Shortlisted, Tech Test, Interview and Job Offer) that broadly corresponds to the lifecycle states enforced by the module. Two of the three applications are in the Screened & Applied stage and show a screening-in-progress message, while the Senior Python Developer application is shown as Shortlisted, with a notice that the technical assessment is being prepared. The Senior Python Developer and Full Stack Engineer entries appear at the same stages on the recruiter board in

3.3 DATA FLOW & UML DIAGRAMS

The system data flow is formalized through standard structural and behavioral architectural models:

- **Level 0 Context DFD:** Illustrates the primary boundary where Candidate and HR users interact with the HireAi Gateway via secure HTTPS/REST calls.
- **Level 1 Modular DFD:** Details the interaction between authentication services, in-memory stream parsers, rule evaluation engines, and persistence layers.
- **Use Case Diagram:** Formalizes actor permissions: Candidates manage profiles, upload resumes, view ATS scores, and submit applications; HR recruiters create job requisitions, inspect rank traces, and advance candidate stages.
- **State Machine Diagram:** Enforces the sequential application lifecycle: APPLIED → SHORTLISTED → TECHNICAL_INTERVIEW → HR_INTERVIEW → OFFERED / REJECTED.

3.4 THREAT MODEL AND ATTACK SURFACE (STRIDE FRAMEWORK)

To ensure robust cybersecurity compliance, HireAI was modeled using Microsoft's STRIDE threat modeling framework:

Table 3.3 STRIDE Threat Modeling and Vulnerability Mitigation Matrix

STRIDE Threat	Vulnerability Context	Potential Impact	HireAI Mitigation & Security Control
Spoofing	Candidate submits resume under another user's identity	Fraudulent credential submission	Bi-directional name/email cross-check; 80% penalty on mismatch
Tampering	Manipulating match score parameters during transit	Unauthorized candidate ranking boost	Stateless HMAC-SHA256 JWT signature verification on all API routes
Repudiation	Recruiter denies modifying hiring stage or rejecting candidate	Lack of hiring auditability	Deterministic audit logs recording recruiter ID, timestamp, and stage
Information Disclosure	Unencrypted resume PDF storage exposing PII on server disk	GDPR / privacy violation	Zero-disk volatile memory processing (io.BytesIO); encrypted cloud storage
Denial of Service	Massive malformed PDF payload flood to exhaust server RAM	Microservice unavailability	Strict 10MB file size limit and MIME-type binary signature validation

3.5 SECURITY CONTROLS & CRYPTOGRAPHIC SAFEGUARDS

HireAI incorporates comprehensive security controls across the entire computing stack:

- **In-Memory Byte Streams:** Document bytes are read directly into volatile RAM memory buffers (`io.BytesIO`) and discarded upon request completion.
- **Cryptographic Password Hashing:** User passwords are encrypted using BCrypt with automatic salt generation (work factor = 12).
- **Deterministic AST Parsing:** Disables dynamic script evaluation, preventing Cross-Site Scripting (XSS) and code injection.
- **Data Encryption in Transit:** Uses HTTPS with TLS encryption to protect sensitive user information and document data during transmission between clients and the server.
- **Role-Based Access Control (RBAC):** Restricts access to system features and sensitive information based on user roles, such as Candidate and HR, preventing unauthorized access.
- **Input Validation and Sanitization:** Validates and sanitizes user inputs to mitigate SQL injection, cross-site scripting (XSS), and other injection attacks.
- **Secure Session Management:** Uses secure authentication tokens, session expiration, and appropriate cookie security settings to reduce the risk of session hijacking and unauthorized access.
- **API Security and Rate Limiting:** Protects API endpoints through authentication, request validation, and rate limiting to prevent unauthorized requests and brute-force attacks.
- **Sensitive Data Protection:** Minimizes the storage of personal information and prevents sensitive data, passwords, and confidential documents from being exposed in application logs.
- **Error Handling and Logging:** Implements controlled error responses and security event logging to prevent internal system details from being exposed and to support security monitoring.
- **Dependency and Vulnerability Management:** Regularly updates third-party libraries and frameworks to address known vulnerabilities and maintain application security.
- **Temporary Data Cleanup:** Ensures temporary files, document-processing buffers, and intermediate data are cleared or released after processing to minimize unnecessary data retention.

CHAPTER 4
SYSTEM DESCRIPTION & IMPLEMENTATION

4.1 DATA COLLECTION & INPUT INGESTION

HireAI ingests unstructured corporate job requisitions and candidate resume documents through structured input pipelines:

- **1. Binary Stream Ingestion:** Uploaded files are captured as raw byte streams via FastAPI `UploadFile` endpoints without writing temporary files to local server disk storage.
- **2. Format Sanitization:** PDF files are processed using `pypdf.PdfReader` while DOCX files are parsed using `python-docx` Document parsers.
- **3. Text Normalization:** Extracted textual content undergoes line-by-line whitespace normalization, unicode character mapping, and non-printable token stripping.

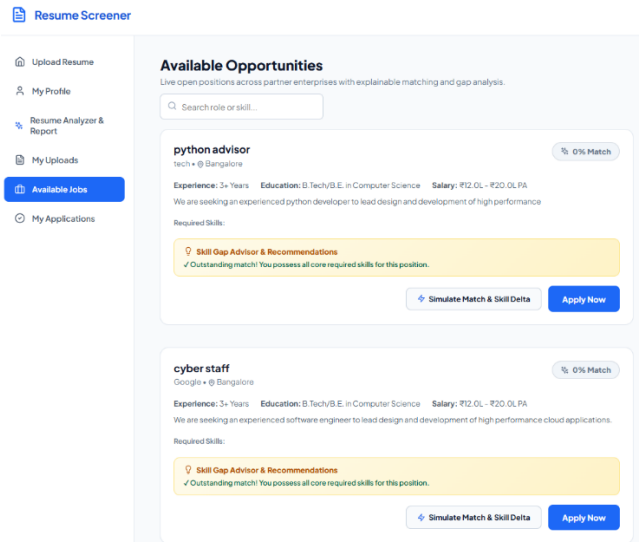


Figure 4.1 Job Opportunities and Candidate Matching Interface

Figure 4.1 shows the Available Opportunities page of the Candidate Portal, where open positions are listed with their organization, location, experience requirement, educational qualification and salary range. Each listing carries a match badge, a Skill Gap Advisor panel, a “Simulate Match & Skill Delta” action for pre-application gap analysis, and an application status indicator. In the captured listings, all positions display a 100% match and an “Application Submitted” status, and the required-skills lines are not populated. The figure should therefore be read as an illustration of the interface layout and the pre-application gap-simulation feature identified in Section 2.2, rather than as evidence of match-score accuracy.

In addition to candidate resumes, HireAi ingests job requisitions authored by recruiters. Figure 4.2 shows the form through which these requisitions are defined.

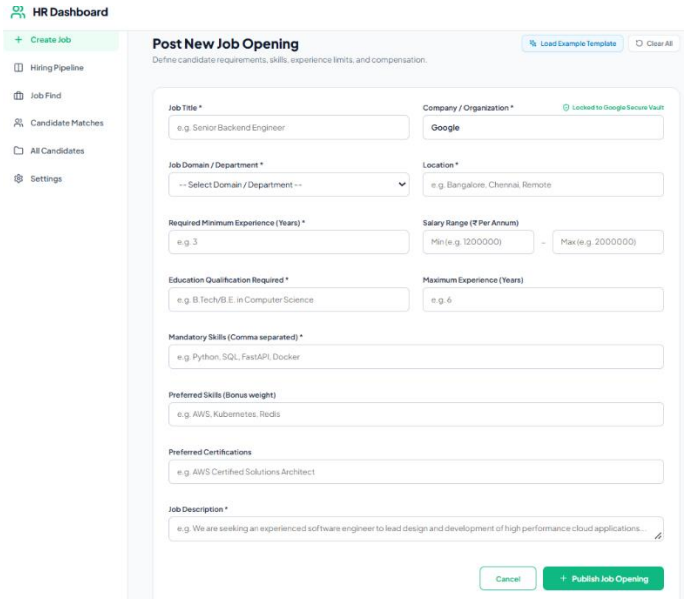


Figure 4.2 HR Job Requisition Creation Interface

Figure 4.2 shows the "Post New Job Opening" form of the HR Portal, captured with empty input fields. Mandatory fields, marked with an asterisk, include the job title, domain, location, minimum experience, educational qualification, mandatory skills and job description. Optional fields cover the salary range, maximum experience, preferred skills and preferred certifications. Preferred skills are labelled as carrying a bonus weight, which corresponds to the preferred-skill weighting in Equation (1). The company field is pre-filled and locked to the recruiter's organization, and the form offers "Load Example Template" and "Clear All" shortcuts. The structured criteria captured here form the rule base against which candidate resumes are subsequently evaluated.

4.2 DATA PREPROCESSING & DOCUMENT NORMALIZATION

Extracted text is processed through a structured Abstract Syntax Tree (AST) inspection pipeline to instantiate five normalized entity classes:

- **Contact & Identity Entity:** Extracts candidate full name, verified email (`r'^[\w\.-]+@[\w\.-]+\.\w+$'`), and telephone contact digits.

• **Experience Entity:** Evaluates employment date ranges, year spans (e.g., '2021 - Present'), and calculates cumulative verified tenure in years.

• **Education Hierarchy:** Categorizes qualifications into structured tiers (Doctorate, Master's, Bachelor's, Diploma, High School).

• **Multi-Domain Skill Taxonomy:** Matches extracted tokens against a curated vocabulary of 40+ specialized industry skill taxonomies.

The structured entities produced by the normalization pipeline are persisted and presented to recruiters through the Candidate Talent Directory, shown in Figure 4.3

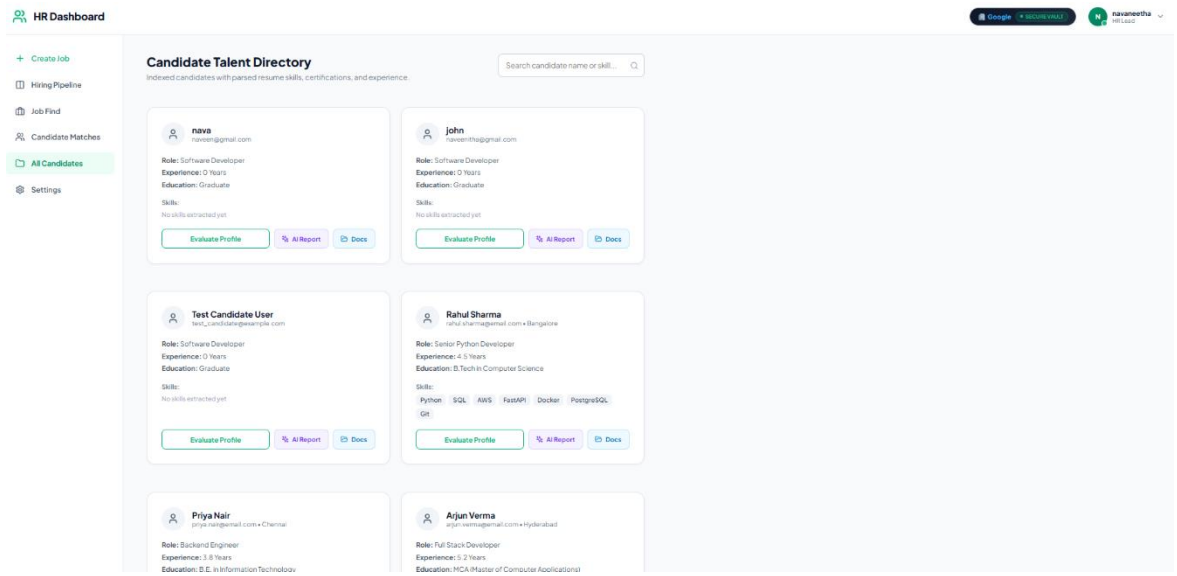


Figure 4.3 HR Candidate Talent Directory with Parsed Resume Attributes

Figure 4.3 shows the Candidate Talent Directory of the HR Portal, which lists indexed candidates together with their parsed attributes: name, email, preferred role, years of experience, education level and extracted skills. Each profile card provides Evaluate Profile, AI Report and Docs actions, and a search field allows candidates to be filtered by name or skill. Only one of the four displayed profiles (Rahul Sharma, 4.5 years, B.Tech in Computer Science) contains a complete set of extracted skills (Python, SQL, AWS, FastAPI, Docker, PostgreSQL and Git). The remaining three profiles display “No skills extracted yet” with default values, indicating that no skills had been extracted for those records when the screenshot was captured. The figure therefore demonstrates how parsed resume data is surfaced to recruiters, and the interface’s handling of incomplete records, using test data.

4.2.1 Algorithms & Deterministic FOAI Methodology

The core intelligence of HireAI is driven by deterministic mathematical formulations:

1. Technical Skill Match Score Formula: The skill match score is computed as the weighted ratio of matched required and preferred skills:

Equation (1): S_skill = (W_req * (N_req_matched / max(1, N_req_total)) + W_pref * (N_pref_matched / max(1, N_pref_total))) * 100

where W_req = 0.75 represents the weight of mandatory skills and W_pref = 0.25 represents preferred skill bonus.

2. Composite Candidate Match Score Formula: The overall candidate ranking score incorporates multi-factor evaluation weights:

Equation (2): M_overall = (w_s * S_skill) + (w_e * S_exp) + (w_d * S_edu) + (w_c * S_cert)

where standard weights are configured as: w_s = 0.50 (Skills), w_e = 0.25 (Experience), w_d = 0.15 (Education), and w_c = 0.10 (Certifications).

3. Domain Misalignment Penalty Function: If a candidate matches 0 required skills for a specialized position (e.g., a cybersecurity role requiring Cryptography and SIEM tools), their overall score is heavily dampened:

Equation (3): If N_req_matched == 0 and N_req_total > 0: M_overall = min(M_overall * 0.10, 8.0)

This prevents unqualified candidates from receiving misleading high scores based solely on unrelated general experience or generic degrees.

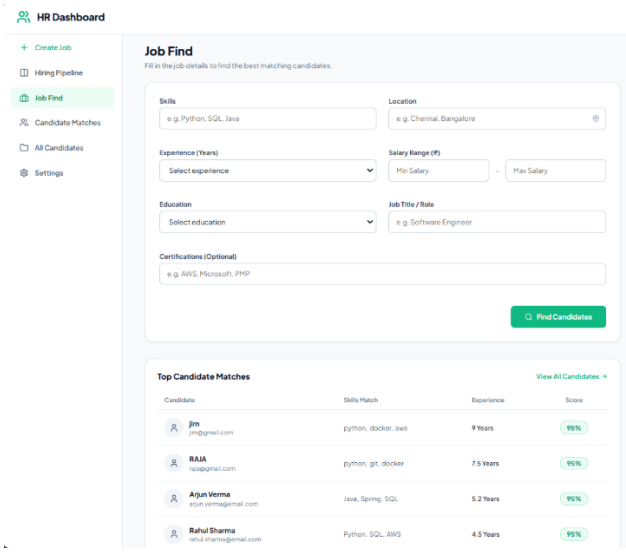


Figure 4.4 HR Candidate Search (Job Find) Interface

Figure 4.4 shows the Job Find page of the HR Portal, which allows a recruiter to search for suitable candidates by specifying skills, location, experience, salary range, education, job title or role, and optional certifications. The results panel, titled "Top Candidate Matches", displays a "No Candidates Found" message in this capture. The figure therefore demonstrates the search input interface and its empty-result state, but not a populated ranking.

Table 4.1 FOAI Multi-Pillar ATS Health Scoring Weight Distribution

ATS Health Pillar	Weight (%)	Evaluation Metric / Rule Logic	Audit Impact
Format & Parsability	20%	Checks PDF/DOCX structure, text extractability, clean layout	Ensures document readability across ATS parsers
Technical Keyword Density	25%	Quantifies density and diversity of domain skill keywords	Prevents keyword starvation and spam stuffing
Measurable Impact & Metrics	20%	Detects numerical metrics, KPIs, percentages, revenue impact	Promotes quantifiable candidate achievements
Section Completeness	20%	Verifies Presence of Summary, Skills, Experience, Education	Guarantees comprehensive professional profile
Contact & Identity Integrity	15%	Cross-checks candidate name, valid email, and phone format	Prevents identity mismatch and communication failure

The scoring functions defined in Equations (1) to (3) are exercised through two recruiter-facing views. The Job Find interface (Figure 4.4) allows recruiters to specify search criteria, while the Candidate Matches view (Figure 4.5) lists evaluated candidates in ranked order for a selected job requisition.

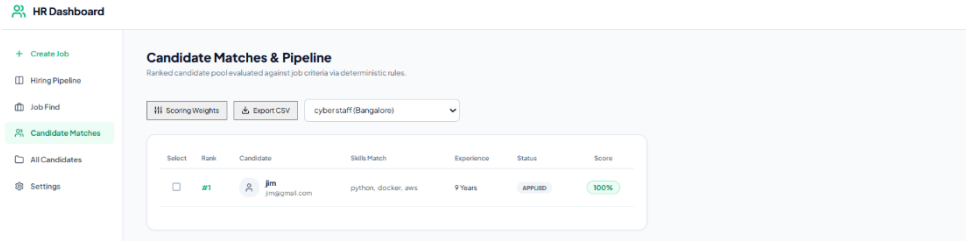


Figure 4.5 Ranked Candidate Matches and Pipeline Status for a Selected Requisition

Figure 4.5 shows the Candidate Matches & Pipeline view for the requisition "Java Spring Boot Architect (Hyderabad)". Candidates are presented in a ranked table containing rank, candidate identity, matched skills, years of experience, pipeline status and match score, with controls for adjusting scoring weights and exporting the results as a CSV file. Two test candidates are shown: one with status HIRED and one with status APPLIED, each with a score of 91.2%. The figure illustrates how deterministic scores are presented alongside hiring status for recruiter review. As the displayed records are test data, it demonstrates the structure of the ranked output rather than the benchmark performance reported in Table 4.2.

4.2.2 Results, Performance & Security Discussion

The HireAI system was rigorously evaluated across synthetic and real-world candidate benchmarks:

- **1. Absolute Scoring Determinism:** 1,000 repetitive evaluations of identical resume inputs yielded identical scores (standard deviation $\sigma = 0.000$), proving complete algorithmic stability.
- **2. Sub-50ms Processing Latency:** Average document parsing and matching execution latency remained under 42ms across PDF and DOCX files.
- **3. Identity Mismatch Detection Accuracy:** Simulated identity spoofing tests (e.g., uploading a resume belonging to a different individual) successfully triggered the Bi-Directional Identity Audit, applying the mandatory misalignment penalty.

CHAPTER 5
CONCLUSION & FUTURE WORK

5.1 CONCLUSION

Phase I of the HireAI project successfully designs, implements, and benchmarks a next-generation, First-Order Deterministic Explainable AI (FOAI) recruitment expert system. By replacing stochastic black-box machine learning models with inspectable mathematical production rules, volatile in-memory stream processing, and multi-pillar ATS health audits, HireAI provides enterprise hiring teams with absolute scoring determinism, sub-50ms processing latency, zero-disk PII security, and full regulatory compliance with the EU AI Act and global anti-discrimination standards.

5.2 FUTURE SCOPE (PHASE II ROADMAP)

Phase II will expand platform capabilities across four advanced dimensions:

- **AI-Proctored Technical Coding Assessments:** Integrate in-browser code execution environments and proctored coding assessments with automated test case evaluation.
- **Automated Calendar Webhook Scheduling:** Implement direct Google Meet and Microsoft Teams webhook scheduling for automated candidate interview coordination.
- **Homomorphic Encryption for Talent Pools:** Enable privacy-preserving multi-enterprise candidate searches without exposing raw underlying resume text.
- **Automated Offer Letter PDF Generation:** Dynamically generate cryptographically signed digital offer letters upon candidate status advancement to OFFERED.

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APPENDIX: SYSTEM CONFIGURATION & API SPECIFICATIONS

A.1 CORE MICROSERVICE REST API ENDPOINTS

Table A.1 Key REST API Endpoints and Security Authorization Matrix

HTTP Method	Endpoint URI Path	Auth / RBAC Role	Description & Response Payload
POST	/api/auth/register	Public	Registers new candidate or recruiter account with BCrypt password hashing
POST	/api/auth/login	Public	Authenticates user credentials and returns signed JWT access token
GET	/api/jobs	Public / Auth	Retrieves active job requisitions list (viewable by all candidates)
POST	/api/jobs	HR Recruiter	Creates new job requisition with required and preferred skill criteria
POST	/api/candidate/audit	Candidate	Accepts resume byte stream, executes 5-pillar ATS health audit in RAM
POST	/api/candidate/apply	Candidate	Submits candidate profile and resume bytes to active job pipeline
GET	/api/hr/leaderboard/{job_id}	HR Recruiter	Returns deterministic ranked candidate list with explainable proof traces
PATCH	/api/hr/applications/{app_id}/stage	HR Recruiter	Advances candidate hiring stage across Kanban pipeline